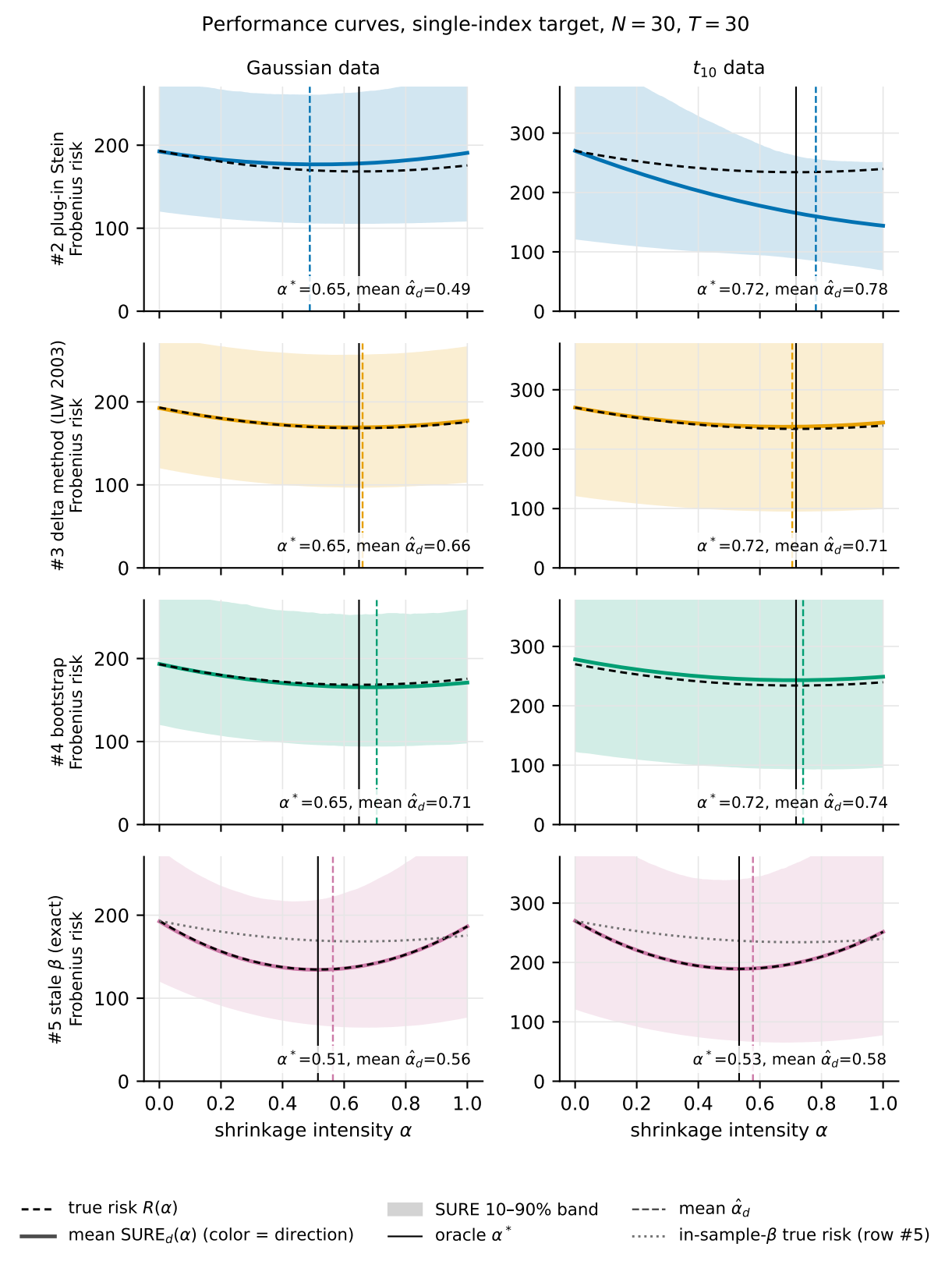

Performance curves, single-index target, $N = 30$, $T = 30$
Gaussian data
$t_{10}$ data
#2 plug-in Stein Frobenius risk
#3 delta method (LW 2003) Frobenius risk
#4 bootstrap Frobenius risk
#5 stale $\beta$ (exact) Frobenius risk
$\alpha^*$=0.65, mean $\hat{\alpha}_d$=0.49
$\alpha^*$=0.72, mean $\hat{\alpha}_d$=0.78
$\alpha^*$=0.65, mean $\hat{\alpha}_d$=0.66
$\alpha^*$=0.72, mean $\hat{\alpha}_d$=0.71
$\alpha^*$=0.65, mean $\hat{\alpha}_d$=0.71
$\alpha^*$=0.72, mean $\hat{\alpha}_d$=0.74
$\alpha^*$=0.51, mean $\hat{\alpha}_d$=0.56
$\alpha^*$=0.53, mean $\hat{\alpha}_d$=0.58
shrinkage intensity $\alpha$
shrinkage intensity $\alpha$
true risk $R(\alpha)$
mean SURE$_d(\alpha)$ (color = direction)
SURE 10–90% band
oracle $\alpha^*$
mean $\hat{\alpha}_d$
in-sample-$\beta$ true risk (row #5)